

CHAPTER 2

Literature review and Project Contribution

Cable-suspended cooperative payload transportation using UAVs has garnered considerable attention in recent years. Researchers have explored a wide range of techniques, from classical methods such as Optimal Control [6] and closed-loop approaches with carrier trajectories calculated online using a Nonlinear Programming (NLP) solver, to learning-based methods such as Imitation Learning and Reinforcement Learning [7] [8] [9], with each offering its own advantages and disadvantages.

2.1 Multi-rotor cooperative transportation

In [6], the authors propose a centralized kinodynamic motion planner that formulates a finite-horizon Optimal Control Problem (OCP) and solves it online in a receding-horizon fashion. The approach aims to enable high-speed, high-acceleration manipulation of a cable-suspended payload while accounting for the strong dynamic coupling between the UAVs, cables, and payload. While it achieves strong results, it relies on a centralized kinodynamic planner, which may pose challenges for fully onboard deployment of the complete control system.

In [7], a centralized kinodynamic planner is distilled using DAgger-based imitation learning into a homogeneous, decentralized UAV policy¹. This allows the approach to achieve agile full-pose cooperative manipulation using only local observations and without communication. While the policy was decentralized, it was not deployed onboard during the actual testing due to its high computational cost.

In [8], a decentralized cooperative control policy was learnt directly through multi-agent reinforcement learning (MAREL), trading explicit trajectory-planning capability for much cheaper computational cost and greater robustness to heterogeneous agents and agent failure. Additionally, the policy was only tasked to learn high level commands/action such as reference acceler-

¹**Distillation** is a form of imitation learning that involves transferring knowledge from a teacher policy, typically obtained through planning or expert demonstration, to a smaller or more deployable student policy by training the latter to reproduce the teacher's actions. **DAgger (Dataset Aggregation)** is an imitation-learning algorithm that iteratively collects expert actions on states visited by the learned policy and aggregates these samples into the training dataset, thereby reducing the distribution-shift problem encountered when the learner visits states not represented in the original demonstrations.

ation. The action is then handed to a low level controller that executes the actual tracking. In this case, since the policy had dramatically lower computational cost, actual onboard execution was possible.

In [9], instead of relying on a low level controller, an end-to-end decentralized cooperative control policy was learnt. The policy takes as input the sensor measurements and outputs the PWM directly to drones' motors. This eliminates the need for a low level controller and simplifies the deployment. However, while the policy did not rely on explicit communication between the drones, it relied on inter-drone measurements, with each drone measuring the relative position between itself and the other drones. Such measurements were subjected to noise during the training to make the policy resilient during deployment in real life scenarios. This is in contrast to [8] where the measurements were in general noise-free but did not employ any type of inter-drone communication or measurements.

2.2 Fixed wings cooperative transportation

However, all of these works have focused on multirotor UAVs, while comparatively little attention has been given to fixed-wing UAVs. More recently, a new line of research has emerged exploring the same problem using fixed-wing UAVs [3] [4] [5]. In [3], the existence of static equilibria for a cable-suspended load transported by non-stopping flying carriers is investigated. The study shows that fixed-wing carriers can maintain a fixed pose for a cable-suspended load using at least three carriers, with some exceptions and degenerate cases. The work presents a theoretical feasibility study supported by numerical simulations in MATLAB/Simulink. However, only steady-state equilibrium is considered, without addressing the transient behavior of the system. Furthermore, the carrier reference trajectories are precalculated offline and executed in open loop, meaning that there is no reaction to disturbances, system uncertainties, or obstacles, and perfect coordination between the carriers is assumed.

In [4], an algorithm for generating coordinated non-stop trajectories for three or more carriers holding a cable-suspended load is presented. The approach uses Hamiltonian cycles and graph theory to generate the coordinated trajectories, with the results demonstrated through numerical simulations in MATLAB/Simulink. However, only steady-state equilibrium is considered, without addressing the transient behavior of the system. Furthermore, the carrier reference trajectories are precalculated and there is no closed-loop control, meaning that the trajectories cannot react to disturbances or system uncertainties during operation.

In [5], a closed-loop control approach for cooperative cable-suspended payload manipulation using non-stopping flying carriers is presented. The approach considers a time-varying load trajectory, which can also include stopping, and consists of an outer-loop load-wrench controller, an inner-loop carrier-trajectory generator, and an optimization layer for the internal forces to enforce a strictly positive carrier velocity norm. The approach is experimentally validated using Crazyflie quadrotors and the Crazyswarm software, with a Vicon motion capture system, in ad-

dition to numerical simulations in MATLAB/Simulink. However, the controller is centralized, is not deployed onboard, and assumes perfect knowledge of the system and clean, noise-free measurements. Furthermore, the trajectories are calculated using a centralized controller together with a Nonlinear Programming (NLP) solver/optimizer, resulting in a high computational cost.

The works discussed above that tackled the non-stopping carriers problems all relied on calculating the UAVs trajectories centrally using classical control techniques. While they demonstrated several effective approaches to cooperative payload transportation using non-stopping carriers, they also highlighted a gap in using learning-based methods and decentralized strategies to tackle the same problem. To the best of the author's knowledge, this research direction was not explored before. Motivated by this gap, this work explores the use of Imitation Learning and Reinforcement learning as a potential research direction.